

Supplementary Material for: Trust-Aware Multi-Axis Reliability Auditing for NLP Classifiers: A Theoretical and Empirical Framework

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This document provides supplementary material organised into seven sections: §1 extended proofs, §2 complete metric definitions, §3 full dataset and preprocessing details, §4 implementation and reproducibility, §5 extended experimental results, §6 additional ablations, and §7 evaluation checklist.

1 Extended Proofs

1.1 Proof of Lemma 1 (ECE controls accuracy–confidence gap)

Lemma 1 (Restated). *Let $\Delta = |\mathbb{E}[\mathbf{1}[\hat{Y} = Y]] - \mathbb{E}[\hat{P}]|$. Then*

$$\Delta \leq \text{ECE} + \sum_{b=1}^B \Pr(\hat{P} \in B_b) |\text{conf}(B_b) - \mathbb{E}[\hat{P}]|.$$

Proof. Partition the probability space into B confidence bins $B_1, \dots, B_B$. Let $w_b = \Pr(\hat{P} \in B_b)$. Then

$$\mathbb{E}[\mathbf{1}[\hat{Y} = Y]] = \sum_{b=1}^B w_b \text{acc}(B_b), \quad (1)$$

$$\mathbb{E}[\hat{P}] = \sum_{b=1}^B w_b \mathbb{E}[\hat{P} \mid \hat{P} \in B_b]. \quad (2)$$

Adding and subtracting $\sum_b w_b \text{conf}(B_b)$:

$$\begin{aligned} \Delta &= \left| \sum_b w_b (\text{acc}(B_b) - \text{conf}(B_b)) + \sum_b w_b (\text{conf}(B_b) - \mathbb{E}[\hat{P} \mid B_b]) \right| \\ &\leq \sum_b w_b |\text{acc}(B_b) - \text{conf}(B_b)| + \sum_b w_b |\text{conf}(B_b) - \mathbb{E}[\hat{P} \mid B_b]|. \end{aligned} \quad (3)$$

The first sum equals ECE by definition. The second sum equals $\sum_b \Pr(\hat{P} \in B_b) |\text{conf}(B_b) - \mathbb{E}[\hat{P}]|$ up to within-bin averaging, yielding the stated bound. $\square$

Remark 1. *The second term vanishes when bin mid-points are used as confidence representatives, recovering the familiar $\Delta \leq \text{ECE}$.*

1.2 Full Proof of Theorem 1 (Pareto Hyperplane)

Theorem 2 (Restated). *Let $m_1, m_2 \in \mathcal{M}$. If $s_a(m_1) \geq s_a(m_2)$ for every axis a with strict inequality on at least one, then $\text{TrustScore}(m_1) > \text{TrustScore}(m_2)$ for every strictly positive α on the simplex. Conversely, every Pareto-optimal trust vector is the unique TrustScore maximiser for some α on the simplex.*

Proof. Forward direction.

$$\text{TrustScore}(m_1) - \text{TrustScore}(m_2) = \sum_a \alpha_a (s_a(m_1) - s_a(m_2)).$$

Each summand is non-negative since $s_a(m_1) \geq s_a(m_2)$; at least one term is strictly positive by the dominance assumption; and $\alpha_a > 0$ for all a . Hence the sum is strictly positive.

Converse direction. Let $\mathbf{t}^* = (s_a(m^*))_a$ be a Pareto-optimal trust vector. Consider the convex hull $\mathcal{C} = \text{conv}\{\mathbf{t}(m) : m \in \mathcal{M}\}$. Since $\mathbf{t}^*$ lies on the Pareto frontier of $\mathcal{C}$, it lies on the boundary of $\mathcal{C}$. By the supporting hyperplane theorem for compact convex sets, there exists a non-zero vector $\alpha \geq \mathbf{0}$ such that

$$\langle \alpha, \mathbf{t}^* \rangle \geq \langle \alpha, \mathbf{t}(m) \rangle \quad \forall m \in \mathcal{M}.$$

Normalising so that $\sum_a \alpha_a = 1$ yields a weight vector on the probability simplex. Pareto-optimality guarantees that no other model dominates m^* , so the inequality is strict for models not on the supporting hyperplane, confirming uniqueness. $\square$

1.3 Proof of Proposition 1 (Boundary Predictor Criterion)

Proposition 1 (Restated). *Let $\pi = \Pr(Y = 1)$ and suppose f_θ outputs the constant prediction $\hat{y}(x) = \mathbf{1}[\pi \geq \frac{1}{2}]$. Then*

$$\text{Acc} = \max(\pi, 1 - \pi), \quad \text{MCC} = 0, \quad \text{F1}_{\text{macro}} = \frac{1}{2}, \quad \text{ECE} \approx |\pi - \frac{1}{2}|.$$

Proof. Without loss of generality assume $\pi \geq \frac{1}{2}$ so the constant prediction is $\hat{y} = 1$.

Accuracy. $\Pr(\hat{Y} = Y) = \Pr(Y = 1) = \pi = \max(\pi, 1 - \pi)$.

MCC. The confusion matrix has $TP = n\pi$, $FP = n(1 - \pi)$, $TN = 0$, $FN = 0$. Then

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} = \frac{0}{\dots} = 0.$$

F1 macro. Class-1 F1 = $2\pi / (2\pi + (1 - \pi)) = 2\pi / (1 + \pi)$; class-0 F1 = 0 (no true negatives predicted). Macro-F1 = $\frac{1}{2} [2\pi / (1 + \pi) + 0]$. At $\pi = 1$ this equals 1; but at $\pi = \frac{1}{2}$ it equals $\frac{1}{2}$. For a constant predictor with no discrimination, $\text{F1}_{\text{macro}} = \frac{1}{2}$ to a first approximation (exact value depends on π).

ECE. The model outputs constant confidence $\hat{p} \equiv \pi$ (or 1 if it predicts the majority class with unit confidence). The single-bin calibration gap is $|\pi - \pi| = 0$ if confidence equals class frequency, but practically $|\hat{p} - \pi|$. When the model is overconfident (output $\hat{p} = 1$), $\text{ECE} = |1 - \pi|$. The approximation $\text{ECE} \approx |\pi - \frac{1}{2}|$ holds when $\hat{p} \approx 1 - \pi + \pi = 1$ and π is close to $\frac{1}{2}$. The bound is tight for the experimental regime $\pi \approx 0.354$. $\square$

1.4 Proof of Theorem 2 (TAMS Optimality Bound)

Theorem 3 (Restated). *Let $\delta(m) = \max(\text{ECE}(m) - \tau_{\text{cal}}, \text{TVD}(m) - \tau_{\text{fair}}, \text{NRD}(m) - \tau_{\text{rob}}, 0)$. For any τ' with $\|\tau' - \tau\|_\infty \leq \epsilon$, the TAMS solution $m_{\tau'}^*$, satisfies $F_1(m_{\tau'}^*) \geq F_1(m_\tau^*) - L\epsilon$, where L is the Lipschitz constant of F_1 over the constraint frontier.*

Proof. For $\|\tau' - \tau\|_\infty \leq \epsilon$, each constraint threshold changes by at most ϵ . Hence the feasible set

$$\mathcal{M}'(\tau) = \{m : \text{ECE}(m) \leq \tau_{\text{cal}}, \text{TVD}(m) \leq \tau_{\text{fair}}, \text{NRD}(m) \leq \tau_{\text{rob}}\}$$

satisfies $\mathcal{M}'(\tau) \subseteq \mathcal{M}'(\tau') \cup S_\epsilon$, where S_ϵ is the ϵ -expansion of the constraint boundary. The optimal m_τ^* may be excluded from $\mathcal{M}'(\tau')$ only if it lies within distance ϵ of a constraint boundary. On this neighbourhood, Lipschitz continuity of F_1 with constant L gives

$$F_1(m_{\tau'}^*) \geq F_1(m_\tau^*) - L\epsilon. \quad \square$$

$\square$

Remark 4. L can be estimated empirically as the maximum pairwise slope $|F_1(m_1) - F_1(m_2)|/\|\mathbf{c}(m_1) - \mathbf{c}(m_2)\|_\infty$ across all model pairs near the boundary, where $\mathbf{c}(m) = (\text{ECE}(m), \text{TVD}(m), \text{NRD}(m))$.

1.5 Proof of Proposition 2 (Optimal Selective Ordering)

Proposition 2 (Restated). *Among all selection functions $c: \mathcal{X} \rightarrow \{0, 1\}$, for any target coverage $\gamma \in (0, 1]$, the minimum selective risk is attained by accepting the γ fraction with the highest oracle correctness probability $\Pr(\hat{Y} = Y | X = x)$.*

Proof. The selective risk at coverage γ is

$$R(\gamma) = \frac{1}{\gamma n} \sum_{i: c(x_i)=1} 1[\hat{y}_i \neq y_i].$$

Minimising $R(\gamma)$ over selection sets of size $\lfloor \gamma n \rfloor$ is equivalent to choosing the $\lfloor \gamma n \rfloor$ samples with lowest individual error probability, i.e., highest $\Pr(\hat{Y} = Y | X = x_i)$. Since confidence $\hat{p}(x_i)$ approximates this probability under calibration, sorting by $\hat{p}$ yields the optimal ordering in the calibrated limit. $\square$ $\square$

2 Complete Metric Definitions

Table 1 provides the complete formal definition of every metric used in TNMM++, together with its direction, domain, and theoretical grounding.

2.1 Axis-to-Trust-Score Normalisation

Each raw metric $v_a(m)$ is normalised to $s_a(m) \in [0, 1]$ as follows:

$$s_a(m) = \begin{cases} \frac{v_a(m) - v_a^{\min}}{v_a^{\max} - v_a^{\min}} & \text{if axis is higher-is-better,} \\ 1 - \frac{v_a(m) - v_a^{\min}}{v_a^{\max} - v_a^{\min}} & \text{if axis is lower-is-better,} \end{cases} \quad (5)$$

where min/max are taken over the evaluated model set $\mathcal{M}$. For the Robustness axis, NRD and NBS are aggregated as $t_{rob} = \frac{1}{2}(1 - \text{NRD}_{\text{norm}} + \text{NBS}_{\text{norm}})$. For the Safety axis, $t_{saf} = \text{SafetyTradeoff}$ if evaluated; otherwise imputed from PR-AUC when prevalence is extreme.

3 Dataset and Preprocessing Details

3.1 DS1: LLM Response Classification

The primary dataset (HuggingFace: `llm-classification-finetuning`) contains 57,477 labelled examples of LLM-generated versus human-written text. Preprocessing steps:

1. Unicode normalisation (NFC).
 2. Removal of HTML markup and non-printable characters.
 3. Truncation to 256 tokens (model tokeniser).
 4. Stratified 90/10 split preserving class frequency ($\pi \approx 0.354$).
- The corruption channel $\mathcal{C}$ applies four perturbations independently with probability $p = 0.1$ per token:
- **Typo injection:** random character swap/insertion/deletion.
 - **Casing flip:** uniform random upper/lower conversion.
 - **Whitespace perturbation:** double-space or space removal.
 - **Synonym substitution:** WordNet first-sense synonym (content words only).
- The normalisation operator $\mathcal{N}$ applies dictionary-based spelling correction using the 6 normalisation rules from DS4, followed by casing and whitespace standardisation.

3.2 DS2: Jailbreak Detection

`biaa-and-jailbreakbench-dataset` provides 900 samples: 884 benign (C0) and 16 adversarial (C1), yielding an extreme imbalance ratio of 55.25. The jailbreak detector g_ψ is a binary MLP trained on token-level features extracted from the frozen encoder. Due to severe class imbalance, the detector is trained with SMOTE oversampling and evaluated threshold-free (PR-AUC) alongside fixed-threshold metrics.

3.3 DS5: CivilComments-WILDS

The CivilComments-WILDS dataset (?) provides demographic group annotations across eight identity categories: *male, female, LGBTQ, Christian, Muslim, other religions, Black, White*. We use the standard WILDS train/ID-test/OOD-test splits. For fairness evaluation, we report the worst-group accuracy and mean parity difference (MPD) averaged over binary group membership indicators. Models are fine-tuned on the WILDS training split and evaluated zero-shot to measure transferability.

3.4 DS6: MultiNLI

MultiNLI dev-matched and dev-mismatched splits are used for zero-shot cross-task generalisation. The task is three-way NLI (entailment / neutral / contradiction); we binarise to (entailment vs. other) for compatibility with the binary classification head. Labels: C0 = entailment, C1 = neutral+contradiction (IR ≈ 1.99).

4 Implementation and Reproducibility

4.1 Software Environment

4.2 Hardware

All experiments were conducted on a single **15.64 GB CUDA GPU** (Colab L4). No multi-GPU or gradient accumulation was used. Memory usage ranged from 4.1 GB (MiniLM) to 13.8 GB (XLM-R) at batch size 32.

4.3 Reproducibility Checklist

- ✓ Fixed random seeds: `torch.manual_seed`, `numpy.random.seed`, `random.seed` set to the run seed before each experiment.
- ✓ Deterministic CUDA operations enabled via `CUBLAS_WORKSPACE_CONFIG=:4096:8`.
- ✓ Checkpoint saved at best val. F1-Macro; final evaluation on fixed validation split.
- ✓ Tokeniser padding/truncation settings: `max_length=256`, `padding='max_length'`, `truncation=True`.
- ✓ Class weights recomputed per-split to prevent leakage.
- × DeBERTa: excluded due to single-device memory constraints at batch size 32 (peak allocation >15.64 GB).

4.4 Hyperparameter Sensitivity

Table 3 shows that the main results are robust to reasonable variations in learning rate and batch size. mBERT retains the highest TrustScore across all configurations tested.

5 Extended Experimental Results

5.1 Full Per-Model Multi-Dimensional Evaluation

Tables 4–7 report the complete metric suite for each of the five mid-tier models in the same format as Table 5 (MiniLM) in the main paper.

5.2 Cross-Model Metric Summary

Table 8 provides a compact cross-model summary of all metrics enabling direct ranking comparison.

6 Additional Ablation Studies

6.1 Ablation S1: TAMS Threshold Sensitivity

Table 9 reports the TAMS-selected model and achieved $F1_M$ under varying threshold configurations, empirically validating Theorem 2.

The bound of Theorem 2 with $L \approx 0.22$ predicts $\Delta F1 \leq 0.022\epsilon$; empirically the maximum drop is 0.0012 at $\epsilon = 0.020$, confirming that the bound is not tight (conservative).

6.2 Ablation S2: Effect of Corruption Severity

We vary the per-token corruption probability $p \in \{0.05, 0.10, 0.20, 0.30\}$ and report accuracy degradation relative to clean for mBERT and MiniLM.

MiniLM shows notably stable accuracy under corruption, consistent with its small hidden dimension (384) inducing lower surface-level feature sensitivity.

Table 1: Complete metric reference for TNMM++. B = number of calibration bins (default 10); n = sample size; k = abstention rank; $g/\bar{g}$ = proxy group and its complement; τ = operating threshold; H = entropy.

Axis	Metric	Dir.	Definition	Domain
Utility	Accuracy	$\uparrow$	$\frac{1}{n} \sum_i \mathbf{1}[\hat{y}_i = y_i]$	$[0, 1]$
	Balanced Acc.	$\uparrow$	$\frac{1}{2}(\text{TPR} + \text{TNR})$	$[0, 1]$
	F1 Macro	$\uparrow$	$\frac{1}{ \mathcal{C} } \sum_c F_1^{(c)}$	$[0, 1]$
	F1 Weighted	$\uparrow$	$\sum_c \frac{n_c}{n} F_1^{(c)}$	$[0, 1]$
	MCC	$\uparrow$	$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	$[-1, 1]$
Calibration	ECE	$\downarrow$	$\sum_{b=1}^B \frac{ B_b }{n} \text{acc}(B_b) - \text{conf}(B_b) $	$[0, 1]$
	MCE	$\downarrow$	$\max_b \text{acc}(B_b) - \text{conf}(B_b) $	$[0, 1]$
	Brier	$\downarrow$	$\frac{1}{n} \sum_i (p_\theta(y_i x_i) - \mathbf{1}[y_i = 1])^2$	$[0, 1]$
	NLL	$\downarrow$	$-\frac{1}{n} \sum_i \log p_\theta(y_i x_i)$	$[0, \infty)$
Robustness	NRD	$\downarrow$	$\frac{A_{clean} - A_{corr}}{\max(A_{clean}, \epsilon)}$	$\mathbb{R}$
	NBS	$\uparrow$	$A_{norm} - A_{corr}$	$[-1, 1]$
Safety	BFPR@ τ	$\downarrow$	$\Pr(g_\psi(x) \geq \tau \mid x \sim \mathcal{D}_{ben})$	$[0, 1]$
	TPR@ τ	$\uparrow$	$\Pr(g_\psi(x) \geq \tau \mid x \sim \mathcal{D}_{jb})$	$[0, 1]$
	SafetyTradeoff	$\uparrow$	$\text{TPR}_\tau(1 - \text{BFPR}_\tau)$	$[0, 1]$
Fairness	TVD(g)	$\downarrow$	$\frac{1}{2} \sum_{c \in \{0,1\}} \pi_g(c) - \pi_{\bar{g}}(c) $	$[0, 1]$
Sel. Pred.	AURC	$\downarrow$	$\int_0^1 \text{risk}(\gamma) d\gamma$	$[0, 1]$
	E-AURC	$\downarrow$	$\text{AURC} - \text{AURC}^*$	$[0, \infty)$
Explainability	Confidence entropy	$\sim$	$\frac{1}{n} \sum_i H(p_\theta(\cdot x_i))$	$[0, \log \mathcal{C}]$
	Mean max conf.	$\uparrow$	$\frac{1}{n} \sum_i \max_c p_\theta(c x_i)$	$[0, 1]$
	Confidence std	$\sim$	$\text{std}_x(\max_c p_\theta(c x))$	$[0, \infty)$

Table 2: Complete software stack used in all experiments.

Component	Version
Python	3.12.12
PyTorch	2.9.0+cu126
Transformers	5.2.0
Datasets	3.6.0
scikit-learn	1.6.1
NumPy	2.0.2
SciPy	1.15.3
NLTK (WordNet)	3.9.1
CUDA	12.6

6.3 Ablation S3: Variance-Adaptive vs. Uniform Weighting

Table 11 compares the top-3 ranking under uniform $\alpha = (1/6, \dots, 1/6)$ versus variance-adaptive α^* from Eq. (7).

The top-ranked model is identical under both schemes; positions 2 and 3 are swapped (margin < 0.002). This confirms that variance-adaptive weighting does not fundamentally alter the ranking while removing manual tuning.

6.4 Ablation S4: Calibration Binning Sensitivity

ECE is sensitive to the number of bins B . Table 12 reports ECE for mBERT and MiniLM across $B \in \{5, 10, 15, 20\}$ using equal-width binning.

Relative ordering is invariant to B ; absolute ECE grows slightly with B due to sparser bins, consistent with known ECE bias (Guo et al., 2017).

Table 3: Hyperparameter sensitivity for mBERT (TrustScore and $F1_M$). Default: $\eta = 2 \times 10^{-5}$, BS=32.

η	BS	$F1_M \uparrow$	Trust $\uparrow$
1×10^{-5}	16	0.5389	0.6631
1×10^{-5}	32	0.5401	0.6648
2×10^{-5}	32	0.5441	0.6685
3×10^{-5}	32	0.5428	0.6672
5×10^{-5}	32	0.5311	0.6554
2×10^{-5}	64	0.5432	0.6677

7 TNMM++ Evaluation Checklist

Table 13 provides a deployment-ready checklist for applying TNMM++ to a new classifier, mapping each evaluation step to the relevant section, metric, and pass/fail threshold.

7.1 Application Protocol

- Data audit.** Verify train/validation/test splits are disjoint and compute class imbalance ratio. Apply class-weighted loss if $\text{IR} > 1.5$.
- Corruption channel.** Construct $\mathcal{D}_{corr}$ and $\mathcal{D}_{norm}$ from the held-out validation set. Do not use corrupted views for checkpoint selection.
- Metric computation.** Compute all checklist metrics in Table 13 on the held-out set.
- Boundary-predictor check.** Apply Proposition 1 criterion jointly across Acc, MCC, ECE. Any model triggering all three conditions is flagged as a degenerate predictor.
- Safety evaluation.** If prevalence $\pi < 0.05$, discard fixed-threshold metrics and report PR-AUC as the primary safety metric.

6. **TrustScore computation.** Normalise all axis scores to $[0, 1]$, apply variance-adaptive weights α^* , and compute TrustScore.
7. **TAMS selection.** Set deployment thresholds $\tau_{cal}, \tau_{fair}, \tau_{rob}$ from operational SLAs. Run TAMS (Algorithm 1). If $\mathcal{M}' = \emptyset$, relax thresholds by $+\epsilon$ and re-run.
8. **Stability audit.** Sample 1,000 Dirichlet weight vectors; report rank stability of the selected model. Re-train with ≥ 3 seeds; report mean $\pm$ std.

References

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, Sydney, Australia. PMLR.

Table 4: Full multi-dimensional evaluation of mBERT (2 epochs). [†] lower is better; otherwise higher is better.

Dimension	Metric	Value	Note
Utility	Accuracy	0.5666	Slight majority bias
	Balanced Acc.	0.5496	Near-symmetric recall
	F1 Macro	0.5441	Strong mid-tier result
	F1 Weighted	0.5624	Support-adjusted
	MCC	0.1002	Positive correlation
Calibration	ECE [†]	0.0295	Lowest in cohort
	MCE [†]	0.1801	Tight worst-bin gap
	Brier [†]	0.4712	Below uninformative baseline
	NLL [†]	0.6721	Best log-loss overall
Robustness	Clean Accuracy	0.5666	Reference point
	Corrupted Acc.	0.5541	Moderate degradation
	Normalised Acc.	0.5588	Partial recovery
	NRD	−0.0221	Small relative drop
	NBS	0.0047	Normalisation helps
Safety	BFPR@0.5	0.0000	No false alarms
	TPR@0.5	0.0000	Threshold degeneracy
	Safety Tradeoff	0.0000	See ROC/PR-AUC
Fairness (TVD [†] , proxy groups)			
	Has digits	0.2841	Largest disparity
	Has punctuation	0.1902	Medium disparity
	Text length (high)	0.1744	Medium disparity
	Upper-case ratio	0.0721	Lowest disparity
Explainability			
	Confidence entropy	0.6512	Moderate spread
	Mean max conf.	0.5891	Slightly overconfident
	Confidence std	0.0544	Low variability
Selective Prediction			
	AURC [†]	0.3917	Best AURC overall
	E-AURC [†]	0.2953	Moderate sub-optimality

Table 5: Full multi-dimensional evaluation of ELEC-TRA (2 epochs). [†] lower is better; otherwise higher is better.

Dimension	Metric	Value	Note
Utility			
	Accuracy	0.5652	Balanced prediction
	Balanced Acc.	0.5517	Highest balanced acc.
	F1 Macro	0.5449	Top F1 in cohort
	F1 Weighted	0.5611	Support-adjusted
	MCC	0.1055	Best MCC overall
Calibration			
	ECE [†]	0.0408	Moderate
	MCE [†]	0.2102	Acceptable worst-bin
	Brier [†]	0.4801	Near-informative
	NLL [†]	0.6792	Competitive
Robustness			
	Clean Accuracy	0.5652	Reference
	Corrupted Acc.	0.5438	Notable degradation
	Normalised Acc.	0.5499	Partial recovery
	NRD	−0.0378	Moderate sensitivity
	NBS	0.0061	Normalisation effective
Safety			
	BFPR@0.5	0.0000	Threshold degeneracy
	TPR@0.5	0.0000	See ROC/PR-AUC
	Safety Tradeoff	0.0000	
Fairness (TVD[†], proxy groups)			
	Has digits	0.2913	Highest disparity
	Has punctuation	0.1984	Medium
	Text length (high)	0.1821	Medium
	Upper-case ratio	0.0698	Lowest
Explainability			
	Confidence entropy	0.6634	Slightly diffuse
	Mean max conf.	0.5801	Near-calibrated
	Confidence std	0.0572	Low variability
Selective Prediction			
	AURC [†]	0.3988	Second-best AURC
	E-AURC [†]	0.3024	Slightly above mBERT

Table 6: Full multi-dimensional evaluation of DistilBERT (2 epochs). [†] lower is better; otherwise higher is better.

Dimension	Metric	Value	Note
Utility			
	Accuracy	0.5755	Slight majority lean
	Balanced Acc.	0.5444	Symmetric
	F1 Macro	0.5429	Competitive
	F1 Weighted	0.5712	Support-adjusted
	MCC	0.0976	Positive correlation
Calibration			
	ECE [†]	0.0281	Near-mBERT
	MCE [†]	0.1892	Good worst-bin
	Brier [†]	0.4731	Competitive
	NLL [†]	0.6734	Competitive
Robustness			
	Clean Accuracy	0.5755	Reference
	Corrupted Acc.	0.5605	Minor degradation
	Normalised Acc.	0.5644	Good recovery
	NRD	−0.0261	Low sensitivity
	NBS	0.0039	Normalisation helps
Safety			
	BFPR@0.5	0.0000	Threshold degeneracy
	TPR@0.5	0.0000	
	Safety Tradeoff	0.0000	
Fairness (TVD[†], proxy groups)			
	Has digits	0.3021	Highest disparity
	Has punctuation	0.2213	Medium
	Text length (high)	0.1981	Medium
	Upper-case ratio	0.0841	Lowest
Explainability			
	Confidence entropy	0.6588	Moderate spread
	Mean max conf.	0.5843	Near-calibrated
	Confidence std	0.0591	Low variability
Selective Prediction			
	AURC [†]	0.3855	Best AURC selectively
	E-AURC [†]	0.2891	Lowest E-AURC overall

Table 7: Full multi-dimensional evaluation of ALBERT (2 epochs). [†] lower is better; otherwise higher is better.

Dimension	Metric	Value	Note
Utility			
	Accuracy	0.5713	Moderate
	Balanced Acc.	0.5233	Weakest balanced acc.
	F1 Macro	0.5234	Lowest in cohort
	F1 Weighted	0.5671	Support-inflated
	MCC	0.0712	Weak correlation
Calibration			
	ECE [†]	0.0584	Moderate
	MCE [†]	0.2441	Larger worst-bin gap
	Brier [†]	0.4891	Near-uninformative
	NLL [†]	0.6841	Slightly elevated
Robustness			
	Clean Accuracy	0.5713	Reference
	Corrupted Acc.	0.5404	Notable degradation
	Normalised Acc.	0.5441	Minimal recovery
	NRD	−0.0541	Highest sensitivity
	NBS	0.0037	Marginal recovery
Safety			
	BFPR@0.5	0.0000	Threshold degeneracy
	TPR@0.5	0.0000	
	Safety Tradeoff	0.0000	
Fairness (TVD [†] , proxy groups)			
	Has digits	0.3312	Highest disparity
	Has punctuation	0.2441	Medium-high
	Text length (high)	0.2109	Medium
	Upper-case ratio	0.0971	Lowest
Explainability			
	Confidence entropy	0.6812	Diffuse dist.
	Mean max conf.	0.5541	Lower confidence
	Confidence std	0.0642	Slightly variable
Selective Prediction			
	AURC [†]	0.4211	Weakest AURC
	E-AURC [†]	0.3247	Weakest E-AURC

Table 8: Cross-model metric summary for all eight models. Arrows indicate direction. **Bold** = best in column; underline = worst. RoBERTa and XLM-R are excluded from Trust columns (boundary-predictor regime).

Model	Acc $\uparrow$	F1 $_M$ $\uparrow$	MCC $\uparrow$	ECE $\downarrow$	Brier $\downarrow$	NRD $\downarrow$	TVD $_{\max}$ $\downarrow$	AURC $\downarrow$	Trust $\uparrow$	Rank
RoBERTa	0.6508	<u>0.3942</u>	<u>0.0000</u>	<u>0.1465</u>	0.4980	-0.035				—
XLM-R	0.6508	<u>0.3942</u>	<u>0.0000</u>	<u>0.1466</u>	0.4981	0.045				—
ELECTRA	0.5652	0.5449	0.1055	0.0408	0.4801	-0.038	0.2913	0.3988	0.6630	2
mBERT	0.5666	0.5441	0.1002	0.0295	0.4712	-0.022	0.2841	0.3917	0.6685	1
DistilBERT	0.5755	0.5429	0.0976	0.0281	0.4731	-0.026	0.3021	0.3855	0.6627	3
MobileBERT	0.5525	0.5372	0.0891	0.0200	0.4631	0.019	0.3142	0.4101	0.6620	4
ALBERT	0.5713	<u>0.5234</u>	0.0712	0.0584	0.4891	<u>-0.054</u>	<u>0.3312</u>	<u>0.4211</u>	<u>0.6354</u>	5
MiniLM †	0.5945	0.5438	0.0889	0.0604	0.4923	0.008	0.3107	0.4099	0.3650	6

Trained for 3 epochs; Trust penalty due to extreme safety axis failure.

Table 9: TAMS-selected model under varying thresholds. Default: $\tau_{cal} = 0.05$, $\tau_{fair} = 0.35$, $\tau_{rob} = 0.06$. ϵ = max deviation from default.

ϵ	Selected	F1 $_M$
0.000	mBERT	0.5441
0.005	mBERT	0.5441
0.010	mBERT	0.5441
0.015	mBERT	0.5441
0.020	ELECTRA	0.5449
0.030	ELECTRA	0.5449
0.050	DistilBERT	0.5429

Table 10: Accuracy under varying corruption severity (p = per-token corruption probability).

Model	$p = 0.05$	$p = 0.10$	$p = 0.20$	$p = 0.30$
mBERT	0.5609	0.5541	0.5401	0.5214
MiniLM	0.5921	0.5896	0.5831	0.5744
ELECTRA	0.5589	0.5438	0.5289	0.5101
DistilBERT	0.5699	0.5605	0.5488	0.5312

Table 11: Top-3 model ranking under uniform vs. variance-adaptive weighting.

Rank	Uniform	Score	Adaptive	Score
1	mBERT	0.6612	mBERT	0.6685
2	DistilBERT	0.6598	ELECTRA	0.6630
3	ELECTRA	0.6587	DistilBERT	0.6627

Table 12: ECE sensitivity to number of bins B .

Model	$B = 5$	$B = 10$	$B = 15$	$B = 20$
mBERT	0.0281	0.0295	0.0304	0.0311
MiniLM	0.0571	0.0604	0.0619	0.0628
ELECTRA	0.0389	0.0408	0.0421	0.0431
DistilBERT	0.0268	0.0281	0.0289	0.0296

Table 13: TNMM++ deployment evaluation checklist. Pass thresholds are indicative and should be adjusted to deployment context. All thresholds assume a binary classification setting with moderate class imbalance ($IR \lesssim 5$).

#	Axis	Metric	Interpretation	Pass Threshold	Ref.
1	Utility	MCC	Checks class-balanced discrimination	> 0.10	§3.3
2	Utility	F1 Macro	Unweighted per-class quality	> 0.50	§3.3
3	Utility	Balanced Acc.	Mean per-class recall	> 0.50	§3.3
4	Calibration	ECE	Avg. confidence gap	< 0.10	Lem. 1
5	Calibration	Brier	Below uninformative baseline	< 0.50	§3.3
6	Calibration	NLL	Log-loss on test set	< 0.70	§3.3
7	Robustness	NRD	Accuracy drop under corruption	< 0.05	§3.2
8	Robustness	NBS	Normalisation recovery	> 0.00	§3.2
9	Safety	PR-AUC	Threshold-free jailbreak detection	> 0.10	§4.3
10	Safety	ROC-AUC	Ranking quality for safety	> 0.65	§4.3
11	Fairness	TVD (max)	Worst-group prediction disparity	< 0.25	Lem. 2
12	Fairness	TVD (mean)	Average proxy-group disparity	< 0.15	Lem. 2
13	Sel. Pred.	E-AURC	Suboptimality of abstention ordering	< 0.35	Prop. 2
14	Boundary test	MCC + ECE + Acc	Joint boundary-predictor signature	Fail: $MCC=0$ & $ECE \approx \pi - \frac{1}{2} $	Prop. 1
15	Composite	TrustScore	Pareto-consistent aggregate	> 0.65	Thm. 1
16	Selection	TAMS feasibility	At least one model satisfies C4, C11, NRD	non-empty $\mathcal{M}'$	Thm. 2